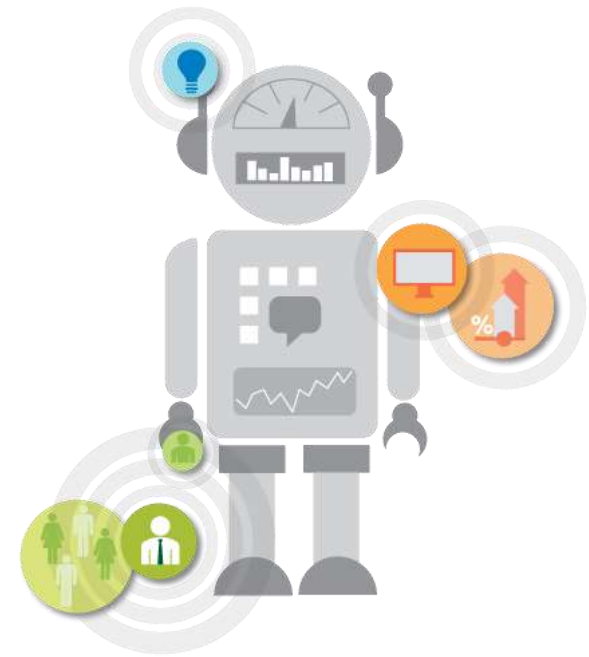


AI and Society

CSET-485

Dr. Pulkit Dwivedi



UNIT - 3



Responsible AI: Risks, Attacks, Governance & Societal Impact

Introduction to AI Risks

- Artificial Intelligence systems are increasingly being deployed in critical domains such as healthcare, finance, governance, and autonomous systems.
- These systems are capable of making decisions that directly impact human lives.
- While AI offers significant benefits in terms of efficiency and automation, it also introduces several risks. These risks arise due to data dependency, model complexity, and lack of interpretability.
- AI systems are often probabilistic, meaning they may produce incorrect or biased outputs. Additionally, large-scale deployment amplifies the consequences of errors.
- Therefore, it becomes essential to study the risks associated with AI systems. Understanding these risks forms the foundation for developing responsible and trustworthy AI.

Attacks on AI Systems

- AI systems are vulnerable to various types of attacks that exploit weaknesses in data, models, or deployment environments.
- These attacks can compromise the reliability and security of AI systems.
- The major categories include adversarial attacks, physical attacks, and membership inference attacks.
- Adversarial attacks manipulate input data to deceive the model. Physical attacks involve real-world modifications to fool AI systems.
- Membership inference attacks target the privacy of training data.
- These attacks highlight that AI systems are not inherently secure. Ensuring robustness against such attacks is critical for safe deployment.
- Therefore, security is a fundamental component of responsible AI.

Adversarial Attacks

- Adversarial attacks involve making small, carefully designed changes to input data to mislead AI models.
- These changes are often imperceptible to humans but significantly affect model predictions. For example, adding slight noise to an image can cause a model to misclassify it.
- These attacks exploit vulnerabilities in machine learning models.
- They are particularly dangerous in safety-critical applications such as autonomous driving and medical diagnosis.
- Adversarial examples reveal that high accuracy does not guarantee robustness. Researchers are developing defense mechanisms such as adversarial training.
- However, complete protection remains a challenge. This makes adversarial robustness an active research area.

Physical Attacks on AI

- Physical attacks occur in real-world environments where objects are modified to deceive AI systems. For instance, adding stickers to a stop sign can cause a self-driving car to misinterpret it.
- These attacks demonstrate that models trained in controlled environments may fail in real-world conditions.
- Physical attacks are harder to detect because they occur outside the digital domain.
- They expose the gap between model training and deployment environments.
- Such attacks can lead to serious safety hazards.
- Therefore, real-world testing is essential for AI systems. Robust design must consider environmental variations.
- Physical robustness is crucial for practical AI deployment.

Membership Inference Attacks

- Membership inference attacks aim to determine whether a particular data point was used in training a model. This can lead to serious privacy violations.
- For example, an attacker may infer whether a patient's medical data was part of a training dataset. These attacks exploit overfitting in machine learning models.
- Even without access to model internals, attackers can use output probabilities to infer membership.
- This poses risks in sensitive domains like healthcare and finance.
- Techniques such as differential privacy can mitigate these risks.
- Proper data anonymization is also important. Ensuring privacy protection is a key aspect of responsible AI.

Need for Responsible AI

- Responsible AI is essential to ensure that AI systems are safe, fair, and trustworthy.
- As AI systems influence critical decisions, their impact on society increases significantly. Without proper safeguards, AI can lead to bias, discrimination, and privacy violations.
- Responsible AI aims to align AI systems with ethical and societal values. It promotes transparency, accountability, and inclusivity.
- It also ensures that AI systems do not harm individuals or communities.
- Governments and organizations are increasingly focusing on ethical AI frameworks.
- Responsible AI builds trust among users and stakeholders. It is crucial for sustainable AI adoption.

Understanding AI System Functioning

- Understanding how AI systems function is critical for safe and reliable deployment.
- Many AI models, especially deep learning systems, operate as black boxes.
- This means their decision-making process is not easily interpretable.
- Lack of understanding can lead to unexpected behavior in real-world scenarios.
- Models may learn spurious correlations instead of meaningful patterns. This can result in incorrect predictions during deployment.
- Explainability techniques help in understanding model decisions.
- Transparent models are easier to debug and improve. Therefore, interpretability is essential for trustworthy AI systems.

Post-Deployment & Stakeholders

- After deployment, AI systems interact with multiple stakeholders including users, developers, regulators, and affected individuals.
- Each stakeholder has different expectations from the system.
- Users require understandable explanations of decisions.
- Regulators need transparency for compliance and auditing.
- Developers need insights for debugging and improvement.
- Lack of explanation reduces trust and adoption.
- Post-deployment monitoring is essential to ensure consistent performance. Feedback mechanisms help improve system reliability. Therefore, stakeholder awareness is critical in AI deployment.

Decision-Making in AI

- AI systems are increasingly used for automated decision-making in areas such as credit scoring, hiring, and healthcare.
- These decisions can have significant consequences on individuals.
- Unlike human decisions, AI decisions may lack contextual understanding. Users often expect fairness and justification for decisions.
- However, black-box models make it difficult to provide explanations. This raises concerns about transparency and trust.
- Decision-making must be explainable and accountable.
- Ethical considerations must be integrated into decision processes. Responsible AI ensures fair and justified decisions.

Consistency Across Stakeholders

- AI systems must provide consistent outcomes across different groups of stakeholders. However, bias in training data can lead to inconsistent results.
- This may result in discrimination against certain groups.
- Bias can arise from data collection, model design, or deployment context.
- Large-scale deployment amplifies these inconsistencies. Ensuring fairness requires careful data selection and model evaluation.
- Techniques such as bias mitigation and fairness metrics are used. Consistency is essential for maintaining social trust.
- Responsible AI promotes equitable outcomes for all stakeholders.

Incorrect Decisions & Exclusion

- AI systems are not always accurate and may produce incorrect decisions. In critical applications, such errors can have serious consequences.
- For example, incorrect decisions may deny individuals access to essential services. This can lead to social exclusion and inequality.
- High accuracy does not guarantee fairness or reliability.
- Therefore, human oversight is necessary in critical decisions. Mechanisms for appeal and correction should be implemented.
- AI systems must minimize false negatives and false positives. Responsible deployment ensures that no individual is unfairly excluded.

Accountability of AI Decisions

- Accountability in AI refers to assigning responsibility for decisions made by AI systems.
- AI systems involve multiple stakeholders such as developers, organizations, and users.
- This creates complexity in assigning responsibility. The “many hands problem” makes it difficult to identify who is accountable. Lack of accountability reduces trust in AI systems.
- Legal and regulatory frameworks are required to address this issue.
- Clear roles and responsibilities must be defined. Accountability mechanisms ensure proper governance. Responsible AI emphasizes traceability and responsibility.

Dimensions of AI Accountability

- AI accountability includes multiple dimensions such as transparency, traceability, responsibility, and auditability. Transparency ensures that decisions can be understood.
- Traceability helps track the source of errors. Responsibility assigns ownership of decisions. Auditability enables independent evaluation of systems.
- These dimensions are essential for regulatory compliance. They also improve trust and reliability of AI systems. Organizations must implement accountability frameworks.
- Continuous auditing is necessary for long-term reliability. These dimensions form the backbone of responsible AI governance.

Governance Structures in AI

- AI governance involves policies, regulations, and institutional frameworks for managing AI systems. It ensures that AI systems operate within ethical and legal boundaries.
- Governance structures include regulatory bodies, standards organizations, and internal policies.
- They define rules for data usage, model deployment, and monitoring. Effective governance balances innovation and risk management.
- It also ensures compliance with legal requirements.
- Organizations must adopt governance frameworks for responsible AI. Continuous updates are required as technology evolves. Governance is essential for sustainable AI adoption.

Data, Performance Goals & Metrics

- AI systems rely heavily on data for training and evaluation. Understanding data quality and distribution is crucial for performance.
- Performance goals must be clearly defined before deployment.
- Metrics such as accuracy, precision, recall, and fairness are used for evaluation. However, focusing only on accuracy is insufficient.
- Ethical metrics such as bias and robustness must also be considered.
- Poor data quality can lead to unreliable models.
- Proper data preprocessing and validation are essential.
- Responsible AI requires balanced and meaningful performance evaluation.

Monitoring Plans for AI Systems

- AI systems require continuous monitoring after deployment. Monitoring ensures that the system performs consistently over time.
- Changes in data distribution can affect model performance.
- Feedback loops help in identifying and correcting errors. Monitoring also helps detect security threats and anomalies.
- Regular updates improve system reliability.
- Performance metrics must be tracked continuously.
- Monitoring plans should include human oversight mechanisms. Continuous evaluation is essential for maintaining trust in AI systems.

AI and Data Privacy

- AI systems process large amounts of data, including sensitive personal information. This raises significant privacy concerns. Unauthorized use of data can lead to serious consequences.
- AI models can also infer sensitive information from non-sensitive data.
- This increases the risk of privacy breaches. Data protection laws are essential to safeguard user information.
- Consent-based data collection is a key requirement. Privacy-preserving techniques must be implemented. Responsible AI ensures that user privacy is protected.

AI Policy & Privacy Protection

- Governments and organizations are developing policies to regulate AI systems.
- These policies focus on privacy, security, and ethical usage. Examples include data protection laws and AI guidelines. Policy frameworks help ensure accountability and transparency.
- They also define standards for responsible AI deployment.
- International collaboration is important for global AI governance. Policy documents provide guidelines for safe AI usage.
- Continuous updates are required to address emerging risks. Policy support is essential for responsible AI ecosystems.

Societal Considerations in AI

- AI systems have broader societal impacts beyond individual users.
- They can influence economic structures, social behavior, and public trust.
- AI may amplify existing inequalities if not properly managed.
- It can also be misused for manipulation or misinformation.
- Societal considerations require ethical evaluation of AI applications.
- Policies must address long-term societal impacts.
- Responsible AI ensures that benefits are distributed fairly. Social inclusivity is a key goal of AI deployment.
- Ethical governance is essential for societal well-being.

Principles for Responsible AI

- Responsible AI is guided by principles such as safety, fairness, transparency, and accountability. It also emphasizes inclusivity and non-discrimination.
- Privacy and security are fundamental components. Human-centric design ensures that AI benefits society. These principles help build trust in AI systems.
- They guide the development and deployment of ethical AI.
- Organizations must integrate these principles into their workflows. Continuous evaluation ensures adherence to these principles.
- Responsible AI is essential for sustainable technological progress.